Query: held-out disorder D^* and its simulated phenotype profile P

$P =$ t_1 t_2 t_3 t_4

Naive regime

candidate causal gene g represented by the terms of its associated disorders

from D^*: t_1 t_2 t_3 t_4 → query terms recovered from the gene's own target annotations

from D_1: t_5 t_6

from D_2: t_7 t_8

circular → scores saturate

Leave-target-out regime

candidate causal gene g represented by the terms of its associated disorders

from D^*: ~~t_1~~ ~~t_2~~ ~~t_3~~ ~~t_4~~ → match must come from phenotypic consistency across the gene's other disorders

terms(D^*) excluded

from D_1: t_5 t_6

from D_2: t_7 t_8

non-circular → measures true inference

Leave-target-out representation: terms(g; D^*) = $\bigcup_{D \in D(g),\, D \neq D^*}$ terms(D)